

CYNTHERA

Drug Repurposing AI Report

Drug: Oseltamivir

Disease: Influenza

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Recommendation: UNCERTAIN

Executive Summary

CYNTERA v2.0 analysis of Oseltamivir → Influenza produced a recommendation of 'UNCERTAIN'. Evidence Strength: 98.5% (HIGH) | Mechanistic Plausibility: 34.8% (LOW) | Risk Level: 0.0% (NONE). 9 claim(s) extracted from literature (9 from openalex), 0 contradiction(s) detected. Prior knowledge: regulatory NONE (0%), repurposing EMERGING, mechanistic WEAK, clinical HUMAN_EVIDENCE, knowledge maturity SPECULATIVE. Safety grade: A. 20 mechanistic path(s) traced.

Three-Dimensional Score Assessment

Dimension	Score	Level	Details
Support Score (SS)	0.985	HIGH	58 evidence records
Mechanistic Score (MS)	0.348	LOW	11 pathways
Risk Score (RS)	0.000	NONE	0 failed trials, 0 contradictions

Therapeutic Direction Alignment (Phase 4D)

Overall Directional Alignment: INSUFFICIENT

Directional evidence for primary target(s) (NA) is insufficient or uncharacterized. Mechanistic chain independently validated in DrugMechDB.

Target	Drug Action	Desired Action	Alignment	Evidence Groups
CACNA1C	UNKNOWN	INHIBITION	INSUFFICIENT	0 supp / 0 opp groups
CURATED_PATH_TARGET	UNKNOWN	UNKNOWN	INSUFFICIENT	0 supp / 0 opp groups
NA	UNKNOWN	UNKNOWN	INSUFFICIENT	0 supp / 0 opp groups
PIK3CD, APDS, IMD14, IMD15, IMD14B, P110DELTA, PI3K, PI3KCHIS, P110D	UNKNOWN	INHIBITION	INSUFFICIENT	0 supp / 0 opp groups
XK	UNKNOWN	UNKNOWN	INSUFFICIENT	0 supp / 0 opp groups

Scientific Context — Prior Knowledge

Dimension	Status	Confidence	Evidence
Regulatory	NONE	0%	No ChEMBL indication match found for this disease
Repurposing	EMERGING	50%	Pair-scoped literature exists for this disease
Mechanistic	WEAK	35%	Validated path: Oseltamivir → NA (P03468) [modulates (ChEMBL)] → Influenza Infection (R-HSA-168255)
Clinical	HUMAN_EVIDENCE	80%	20 registered trial(s) for this pair: NCT01050257 (Phase III, COMPLETED)
Knowledge Maturity	SPECULATIVE	0%	No prior knowledge cache entries for this pair

Candidate Biological Mechanisms

Candidate Mechanism 1: Mechanism 1: NA (P03468) via Influenza Infection (R-HSA-168255)

[**WEAK_SPECULATIVE**] [DIR: UNKNOWN]

Drug: Oseltamivir → Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) → Gene: SEC13 → Disease: Influenza

- Drug: Oseltamivir —**MODULATES**→ Target: NA (P03468) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: DIRECTION UNCERTAIN; Source: ChEMBL

- Target: NA (P03468) —**PARTICIPATES_IN**→ Pathway: Influenza Infection (R-HSA-168255) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Influenza Infection (R-HSA-168255) —**CONTAINS_ASSOCIATED_GENE**→ Gene: SEC13 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: SEC13 —**ASSOCIATED_WITH**→ Disease: Influenza [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Influenza Infection (R-HSA-168255) → Gene: SEC13 is not supported by a mapped claim.

Candidate Mechanism 2: Mechanism 2: NA (P03468) via Influenza Infection (R-HSA-168255)

[**WEAK_SPECULATIVE**] [DIR: UNKNOWN]

Drug: Oseltamivir → Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) → Gene: KPNA1 → Disease: Influenza

- Drug: Oseltamivir —**MODULATES**→ Target: NA (P03468) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: DIRECTION UNCERTAIN; Source: ChEMBL

- Target: NA (P03468) —**PARTICIPATES_IN**→ Pathway: Influenza Infection (R-HSA-168255) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Influenza Infection (R-HSA-168255) —**CONTAINS_ASSOCIATED_GENE**→ Gene: KPNA1 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: KPNA1 —**ASSOCIATED_WITH**→ Disease: Influenza [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Influenza Infection (R-HSA-168255) → Gene: KPNA1 is not supported by a mapped claim.

Candidate Mechanism 3: Mechanism 3: NA (P03468) via Influenza Infection (R-HSA-168255) [WEAK_SPECULATIVE] [DIR: UNKNOWN]

Drug: Oseltamivir → Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) → Gene: RANBP2 → Disease: Influenza

- Drug: Oseltamivir —**MODULATES**→ Target: NA (P03468) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)

Status: DATABASE SUPPORTED; Directionality: DIRECTION UNCERTAIN; Source: ChEMBL

- Target: NA (P03468) —**PARTICIPATES_IN**→ Pathway: Influenza Infection (R-HSA-168255) [\[Open Reactome Pathway\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome

- Pathway: Influenza Infection (R-HSA-168255) —**CONTAINS_ASSOCIATED_GENE**→ Gene: RANBP2 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)

Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET

- Gene: RANBP2 —**ASSOCIATED_WITH**→ Disease: Influenza [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)

Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET

Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.

No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Influenza Infection (R-HSA-168255) → Gene: RANBP2 is not supported by a mapped claim.

Candidate Mechanism 4: Mechanism 4: NA (P03468) via Influenza Infection (R-HSA-168255)
[WEAK_SPECULATIVE] [DIR: UNKNOWN]

- Drug: Oseltamivir → Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) → Gene: NUP214 → Disease: Influenza
- Drug: Oseltamivir —**MODULATES**→ Target: NA (P03468) [\[Open ChEMBL Compound\]](#) [\[Open UniProt\]](#)
Status: DATABASE SUPPORTED; Directionality: DIRECTION UNCERTAIN; Source: ChEMBL
 - Target: NA (P03468) —**PARTICIPATES_IN**→ Pathway: Influenza Infection (R-HSA-168255) [\[Open Reactome Pathway\]](#)
Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome
 - Pathway: Influenza Infection (R-HSA-168255) —**CONTAINS_ASSOCIATED_GENE**→ Gene: NUP214 [\[Open Reactome Pathway\]](#) [\[Open Targets Target\]](#)
Status: STRUCTURAL EVIDENCE; Directionality: DIRECTION UNCERTAIN; Source: Reactome + Open Targets/DisGeNET
 - Gene: NUP214 —**ASSOCIATED_WITH**→ Disease: Influenza [\[Open Targets Disease\]](#) [\[Open DisGeNET\]](#)
Status: DATABASE SUPPORTED; Directionality: NOT APPLICABLE; Source: Open Targets / DisGeNET
- Score = 0.20 edge validity + 0.15 directionality + 0.20 disease relevance + 0.15 curated support + 0.25 literature support + 0.05 independence – 0.25 contradiction.

Reactome pathway membership is structural evidence and cannot by itself validate a causal bridge.
No mapped literature claim supports a critical biological bridge; score is capped below moderate support.

Missing evidence: Direct biological bridge for Target: NA (P03468) → Pathway: Influenza Infection (R-HSA-168255) is not supported by a mapped claim.

Missing evidence: Direct biological bridge for Pathway: Influenza Infection (R-HSA-168255) → Gene: NUP214 is not supported by a mapped claim.

Biomedical Data Sources Accessed

Source Name	Status	Direct Source Portal
ChEMBL	SUCCESS	[Open ChEMBL]
UniProt	SUCCESS	[Open UniProt]
Reactome	SUCCESS	[Open Reactome]
Open Targets	SUCCESS	[Open Open Targets]
DisGeNET	SUCCESS	[Open DisGeNET]
PubMed	SUCCESS	[Open PubMed]
Europe PMC	SUCCESS	[Open Europe PMC]
ClinicalTrials.gov	SUCCESS	[Open ClinicalTrials.gov]

Recommendation Rule Engine Output

- Rule 5 (UNCERTAIN): Mixed or sparse evidence. SS=0.985, MS=0.348, RS=0.000. Safety grade: A.
- Evidence signals:
- [PASS] Literature support: SS = 0.985 (HIGH) from 58 records
- [FAIL] Mechanistic plausibility: MS = 0.348 (LOW), 11 pathway(s)
- [PASS] Safety/Risk acceptable: RS = 0.000 (NONE)
- [PASS] Evidence consistency: No contradictions
- [PASS] Human clinical data: Available

Data Gaps Identified

- No prior knowledge entries found for this drug-disease pair. This appears to be a genuinely novel repurposing hypothesis (−0.05 confidence penalty).

Evidence Synthesis & Dimensional Assessment

1. Mechanistic Assessment

Mechanistic Assessment: LOW (Score = 0.348).

Literature Grounding: MODERATE.

Candidate Biological Mechanisms Discovered: 5.

Primary Mechanistic Chain: Drug: Oseltamivir → Target: NA (NA (Neuraminidase) [UniProt: P03468]) [MODULATES via ChEMBL] → Pathway: Influenza Infection (R-HSA-168255) [PARTICIPATES_IN via Reactome] → Gene: SEC13 [CONTAINS_ASSOCIATED_GENE via Reactome + Open Targets/DisGeNET] → Disease: Influenza [ASSOCIATED_WITH via Open Targets / DisGeNET].

Rationale: Mechanistic Score (MS) derived from candidate mechanism quality. Best candidate support level: WEAK_SPECULATIVE. Discovered 5 candidate biological mechanism(s). Literature grounding: MODERATE (9 claim(s)). Final MS = 0.348 (LOW).

2. Clinical Evidence Assessment

Clinical Evidence Assessment: HIGH Literature Support (Score = 0.985).

Clinical Trial Status: RETRIEVED.

Clinical Trial Findings: 20 trial(s) evaluated, 0 clinically significant failure(s).

Literature Support Rationale: Support Score (SS) from 58 evidence records.

Breakdown: Cell-line (in vitro): 2 records (avg ERW: 0.30); Reviews/Observational: 55 records (avg ERW: 0.62); Extracted claims: 1

Formula: $SS = 1 - \exp(-0.12 \times \text{quality_weighted_sum})$. Quality-weighted sum = 34.93 (raw ERW sum would be 36.67, capped to prevent inflation from low-quality records).

Final SS = 0.985 (HIGH).

3. Safety & Risk Assessment

Safety & Risk Assessment: Grade A — Excellent — clean safety record across all trials (Risk Score = 0.000, NONE).

Boxed Warning: NO.

Risk Factors & Signal Analysis: Risk Score (RS) computed from clinically significant safety signals.

Trial analysis: 0 clinically significant failure(s); 0 administrative/external termination(s) excluded.

Risk factors: Safety grade: A (grade penalty: 0.0).

Formula: $RS = 1 - \exp(-0.30 \times \text{raw_risk_burden})$. Raw burden = 0.00. Final RS = 0.000 (NONE).

4. Final Synthesis

Synthesis & Recommendation: UNCERTAIN.

Regulatory Status: NONE (Term: N/A).

1. Mechanistic Plausibility: LOW (0.348)

2. Literature/Clinical Support: HIGH (0.985)

3. Risk & Safety Profile: NONE (0.000, Grade A)

Conclusion Summary: Rule 5 (UNCERTAIN): Mixed or sparse evidence. SS=0.985, MS=0.348, RS=0.000.

Safety grade: A.

Confidence Narrative

Overall confidence is estimated at 95% after accounting for data gaps (total penalty: -5%). Evidence quality: The analysis is based on 58 retrieved records representing moderate-quality mixed evidence. ERW values are capped per evidence tier to prevent inflation from low-quality records (e.g., reviews and hypotheses). Safety assessment: Grade A — Excellent — clean safety record across all trials. No boxed warning detected. Safety grade scale: A = Excellent | B = Acceptable | C = Caution | D = Unsafe. Prior knowledge: No ChEMBL indication data available and no cache entries found for Oseltamivir → Influenza. Treating as a novel repurposing hypothesis. Evidence evaluated on its own merits.